

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Project Planning

ShopEZ MERN E-Commerce Application

Phase: Project Planning Phase

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Project Planning

1. Introduction

This document formalizes the project planning artifacts for the **ShopEZ MERN E-Commerce Application**. It includes the Product Backlog, Sprint Schedule, Sprint Timeline, Velocity Tracking, and Project Progress Analysis. The project follows the Agile Scrum methodology and is executed in three development sprints. The system enables users to browse products, add items to cart, place orders, track purchases, and allows administrators to manage products, inventory, and customer orders efficiently.

2. Product Backlog & Sprint Schedule

SPRINT 1

Duration: Weeks 1-2

Sprint	Epic	USN	User Story / Task	Points	Priority
Sprint 1	User Registration	USN-01	As a new user, I can register using email/phone so that I can create an account.	3	High
Sprint 1	User Authentication	USN-02	As a registered user, I can login securely so that I can access my account.	2	High
Sprint 1	Product Browsing	USN-03	As a shopper, I can browse products by category and view product details so that I can choose items to buy.	5	High
Sprint 1	Search & Filter	USN-04	As a shopper, I can search and filter products by category, price, rating, etc. so that I can find products easily.	3	Medium
Sprint 1	Add to Cart	USN-05	As a shopper, I can add products to my cart so that I can purchase them later.	3	High
Sprint 1	View Cart	USN-06	As a shopper, I can view and update items in my cart so that I can review my order before checkout.	2	High
Sprint 1	Checkout & Order Placement	USN-07	As a shopper, I can place an order by providing shipping details and payment method.	5	High
Sprint 1	Order Confirmation	USN-08	As a shopper, I can receive an order confirmation with order summary and tracking details.	2	Medium

2. Product Backlog & Sprint Schedule (Contd.)

SPRINT 2 Duration: Weeks 3–4

Sprint	Epic	USN	User Story / Task	Points	Priority
Sprint 2	Shopping Cart	USN-09	As a shopper, I can add products to my cart so that I can purchase them later.	2	High
Sprint 2	Shopping Cart	USN-10	As a shopper, I can update quantity or remove products from my cart.	3	High
Sprint 2	Checkout & Payment	USN-11	As a shopper, I can proceed to checkout and make payment using multiple payment methods.	5	High
Sprint 2	Shipping Management	USN-12	As a shopper, I can provide shipping details and choose delivery options.	3	High
Sprint 2	Order Management	USN-13	As a shopper, I can view my order history and order details.	3	High
Sprint 2	Wishlist	USN-14	As a shopper, I can add products to my wishlist and view them later.	2	Medium
Sprint 2	Coupons & Discounts	USN-15	As a shopper, I can apply coupon codes and view available discounts at checkout.	2	Medium
Sprint 2	User Profile Management	USN-16	As a user, I can update my profile information, address, and contact details.	2	Medium

2. Product Backlog & Sprint Schedule (Contd.)

SPRINT 3 Duration: Weeks 5–6

Sprint	Epic	USN	User Story / Task	Points	Priority
Sprint 3	Admin Dashboard	USN-17	As an admin, I can view dashboard with total sales, orders, users, and products summary.	3	High
Sprint 3	Product Management	USN-18	As an admin, I can add new products with name, price, category, and images.	3	High
Sprint 3	Product Management	USN-19	As an admin, I can update or delete existing products.	3	High
Sprint 3	Order Management	USN-20	As an admin, I can view and manage all customer orders and their status.	3	High
Sprint 3	Customer Management	USN-21	As an admin, I can view and manage registered customers.	2	High
Sprint 3	Reports & Analytics Dashboard	USN-22	As an admin, I can view sales reports and analytics for better insights.	3	Medium
Sprint 3	Review & Rating Management	USN-23	As a user, I can write and view product reviews and ratings.	2	Medium
Sprint 3	Performance & Bug Fixing	USN-24	As a developer, I can optimize performance and fix bugs for a smooth user experience.	3	Medium

3. Sprint Timeline — Gantt Chart

Project: Shopez E-Commerce Application

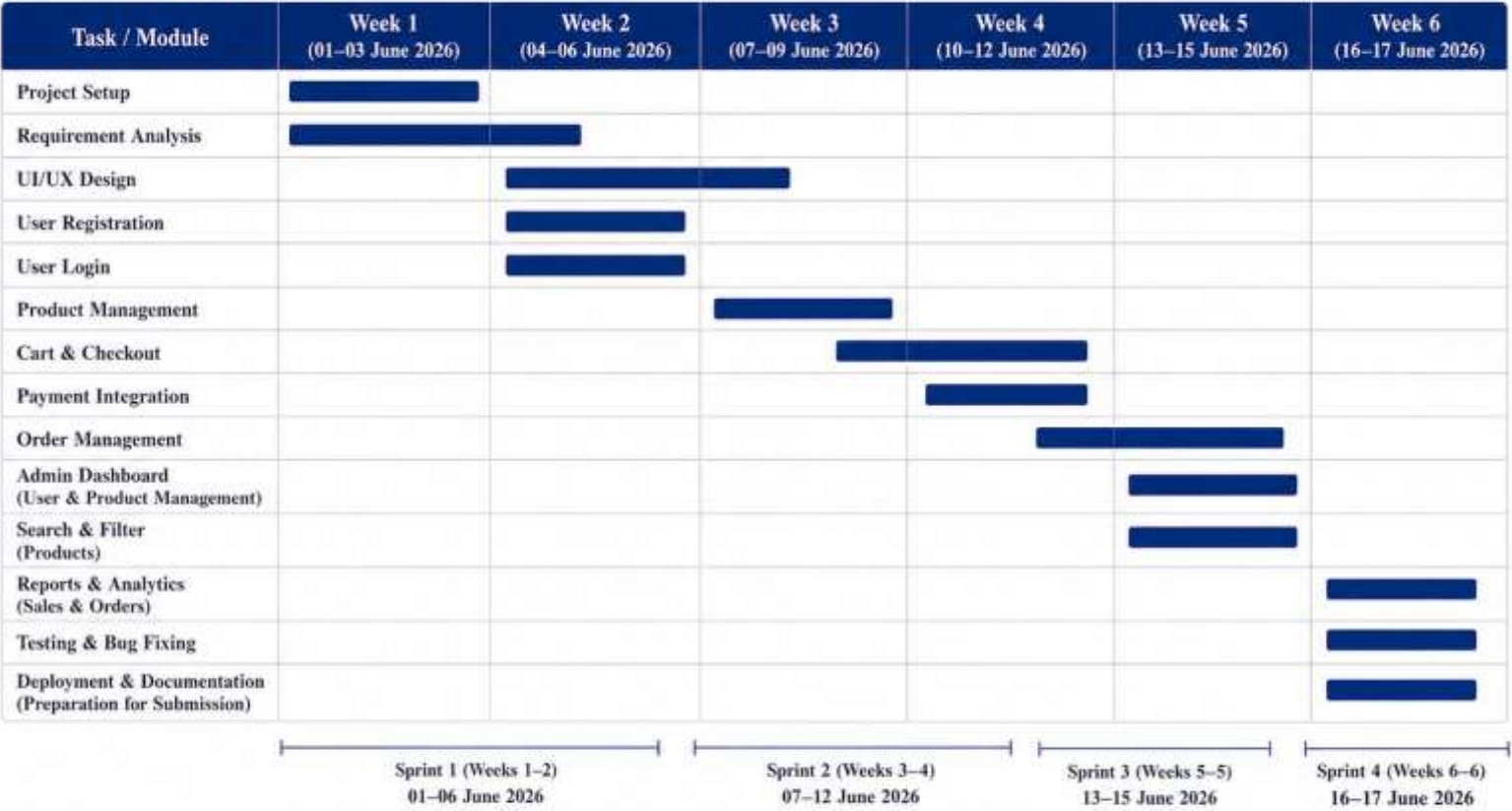


FIGURE 3.1 – Gantt Chart: Shopez E-Commerce Application Development Timeline (4 Sprints × 2–3 Weeks)

4. Velocity & Project Tracker

Sprint	Story Points	Start Date	End Date	Completed	Rate
Sprint 1	10	01 June 2026	06 June 2026	10	100% ✓
Sprint 2	11	07 June 2026	12 June 2026	11	100% ✓
Sprint 3	8	13 June 2026	15 June 2026	8	100% ✓
Total	29	—	—	29	100% ✓

Average Velocity: 9.7StoryPoints per Sprint | Sprint Duration: 2 Weeks

5. Risk Register

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy
R-01	Website unavailable due to server downtime	Medium	High	Deploy on reliable cloud hosting and monitor server health
R-02	MongoDB database connection failure	Low	High	Configure automatic reconnection and maintain database backups
R-03	Unauthorized access to admin dashboard	Medium	Critical	Implement JWT authentication and role-based access control
R-04	Invalid product or order data submitted	Medium	Medium	Apply frontend and backend validation.
R-05	High traffic causing performance issues	Low	Medium	Use pagination, caching, and optimize database queries
R-06	Payment processing failure	Low	High	Implement proper payment gateway validation and error handling

6. Conclusion

The Agile Scrum framework provided a structured and efficient approach for developing the **ShopeZ MERN E-Commerce Application**. The project was successfully completed through planned development sprints covering user authentication, product browsing, shopping cart management, order processing, and administrative product management.

All planned story points were completed within the project timeline, resulting in a fully functional MERN Stack application. The system enables customers to browse products, place orders, and track purchases while allowing administrators to manage inventory and sales efficiently. The final application satisfies the defined functional requirements and is ready for deployment and future enhancements.